SPEECH ENHANCEMENT USING SPECTRAL SUBTRACTION

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ABSTRACT

This project aims to improve the quality of speech signals corrupted by background noise. Spectral subtraction, a frequency-domain technique, is employed to achieve this goal. The method involves estimating the noise spectrum from a noise-only segment of the signal and subsequently subtracting it from the speech spectrum.

The implementation begins by loading a noisy speech signal. A noise-only segment is identified and used to calculate the noise spectrum. The noisy speech signal is then transformed into the frequency domain using the Discrete Fourier Transform (DFT). The estimated noise spectrum is subtracted from the speech spectrum, effectively reducing the noise component. Finally, the enhanced speech signal is obtained by performing the inverse DFT.

To evaluate the effectiveness of the spectral subtraction technique, the original and enhanced speech signals are compared. The project will analyze the reduction in noise and the improvement in speech intelligibility.

Software description

MATLAB (Matrix Laboratory): MATLAB is a high-level programming environment widely used for numerical computation, data analysis, algorithm development, and visualization. Its powerful built-in functions and toolboxes make it ideal for handling complex mathematical operations, such as signal processing, control systems, machine learning, and more. In this project, MATLAB is utilized for implementing a speech enhancement algorithm through spectral subtraction. Its robust support for matrix operations, Fast Fourier Transform (FFT), and audio processing functions, combined with its user-friendly interface for data visualization, makes MATLAB an excellent choice for developing and analyzing signal processing algorithms. The ability to read, manipulate, and output audio data further highlights its flexibility in real-time applications like noise reduction and speech enhancement.

Theory

The project is based on **spectral subtraction**, a widely used technique for speech enhancement and noise reduction. The underlying principle of spectral subtraction involves separating the noise from the noisy speech signal in the frequency domain and reconstructing the clean speech signal by reducing the noise components.

1. Noisy Speech Signal

In a real-world environment, speech signals are often corrupted by background noise. A noisy speech signal y(n)y(n)y(n) can be modeled as:

$$y(n)=x(n)+d(n)$$

Where:

- ***x(n)*** is the clean speech signal.
- ***d(n)*** is the additive noise signal.
- ***y(n)*** is the noisy signal.

The goal of speech enhancement is to recover x(n)from y(n)by reducing the noise d(n).

2. Frame-Based Processing

The noisy signal is processed in **frames** rather than as a continuous stream. This approach enables localized analysis in the time-frequency domain, which is essential for handling non-stationary signals like speech. In this project, the signal is divided into overlapping frames, each of size NNN, with a typical overlap of 50%. Overlapping helps in reducing artifacts during signal reconstruction.

Each frame of the noisy speech signal is windowed using a **Hamming window** to reduce discontinuities at the frame boundaries.

3. Fourier Transform and Frequency Domain Analysis

Once the signal is divided into frames, the **Fast Fourier Transform (FFT)** is applied to convert the signal from the time domain to the frequency domain. This allows the signal to be represented as:

$$Y(k)=X(k)+D(k)$$

Where:

- Y(k), X(k), and D(k) are the Fourier transforms of the noisy signal, clean speech, and noise, respectively.
- k is the frequency bin index.

In the frequency domain, the signal can be expressed as a combination of the magnitude $|Y(k)|$ and the phase $\angle Y(k)$:

$$Y(k)= |Y(k)| e^{(j\angle Y(k))}$$

4. Noise Estimation

An important step in spectral subtraction is the estimation of the noise spectrum. Initially, the magnitude of the first frame is assumed to represent the noise spectrum. For subsequent frames, the noise estimate is updated using a weighted averaging method:

$$\check{D}(k)=\beta\check{D}(k)+(1-\beta) |Y(k)|$$

Where:

- $\check{D}(k)$ is the estimated noise magnitude.

- β is the weighting factor, typically set close to 1 (in this case, 0.9) to give more importance to previous noise estimates.

5. Spectral Subtraction

Once the noise estimate is available, the clean speech spectrum can be estimated by subtracting the noise estimate from the noisy magnitude spectrum:

$$|X(k)| = \max(|Y(k)| - \alpha \check{D}(k), 0)$$

Where:

- α is an over-subtraction factor used to account for errors in noise estimation (usually set greater than 1 to avoid underestimating the noise).
- The subtraction is clipped at 0 to prevent negative magnitudes.

The clean speech spectrum is then reconstructed using the original phase information:

$$X(k) = |X(k)| e^{j\angle Y(k)}$$

6. Inverse FFT and Signal Reconstruction

After estimating the clean speech spectrum in the frequency domain, the **Inverse FFT (IFFT)** is applied to convert the signal back to the time domain. The enhanced frames are then overlapped and added to reconstruct the final enhanced speech signal. Applying a window function (such as Hamming) during both analysis and synthesis helps in minimizing distortions caused by overlapping.

7. Normalization

To avoid any distortion due to amplitude clipping, the enhanced signal is normalized by dividing the entire signal by its maximum absolute value, ensuring the signal's amplitude remains within an acceptable range for playback and further processing.

8. Spectral Subtraction Considerations

While spectral subtraction is effective in reducing noise, there are a few considerations:

- **Musical Noise**: Spectral subtraction can introduce musical noise artifacts due to the random nature of noise, especially if the noise estimate is not accurate.
- **Trade-off**: The over-subtraction factor α needs careful tuning. A larger α reduces more noise but risks removing parts of the speech signal, while a smaller α may leave residual noise.

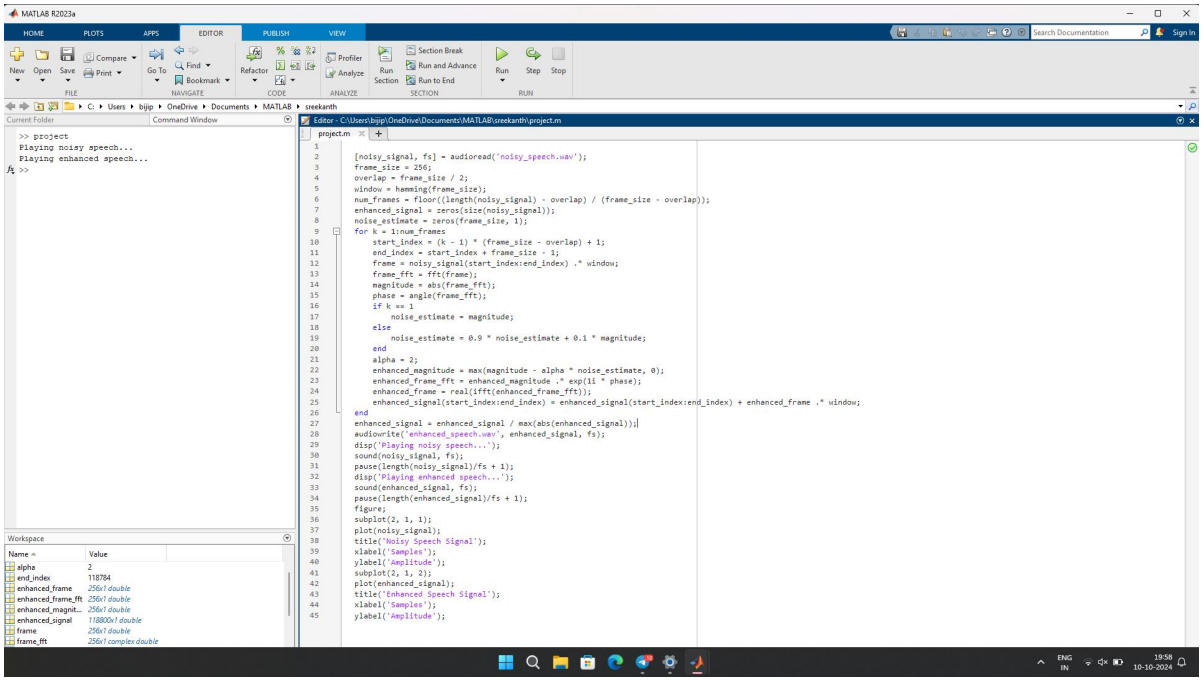
9. Visualization

The noisy and enhanced signals are visualized to compare the effects of noise reduction. The waveforms give a visual representation of how the noise has been reduced and how the overall structure of the speech signal has been preserved.

Code

```
[noisy_signal, fs] = audioread('noisy_speech.wav');
frame_size = 256;
overlap = frame_size / 2;
window = hamming(frame_size);
num_frames = floor((length(noisy_signal) - overlap) /
(frame_size - overlap));
enhanced_signal = zeros(size(noisy_signal));
noise_estimate = zeros(frame_size, 1);
for k = 1:num_frames
   start_index = (k - 1) * (frame_size - overlap) + 1;
   end_index = start_index + frame_size - 1;
   frame = noisy_signal(start_index:end_index) .* window;
   frame_fft = fft(frame);
   magnitude = abs(frame_fft);
   phase = angle(frame_fft);
   if k == 1
      noise_estimate = magnitude;
   else
      noise_estimate = 0.9 * noise_estimate + 0.1 * magnitude;
   end
   alpha = 2;
   enhanced_magnitude = max(magnitude - alpha *
noise_estimate, 0);
   enhanced_frame_fft = enhanced_magnitude .* exp(1i *
phase);
   enhanced_frame = real(ifft(enhanced_frame_fft));
   enhanced_signal(start_index:end_index) =
enhanced_signal(start_index:end_index) + enhanced_frame .*
window;
end
enhanced_signal = enhanced_signal /
max(abs(enhanced_signal));
```

```
audiowrite('enhanced_speech.wav', enhanced_signal, fs);
disp('Playing noisy speech...');
sound(noisy_signal, fs);
pause(length(noisy_signal)/fs + 1);
disp('Playing enhanced speech...');
sound(enhanced_signal, fs);
pause(length(enhanced_signal)/fs + 1);
figure;
subplot(2, 1, 1);
plot(noisy_signal);
title('Noisy Speech Signal');
xlabel('Samples');
ylabel('Amplitude');
subplot(2, 1, 2);
plot(enhanced_signal);
title('Enhanced Speech Signal');
xlabel('Samples');
ylabel('Amplitude');
```



Advantages

1. **Effective Noise Reduction**: The spectral subtraction method efficiently reduces background noise, especially in stationary noise environments like fan noise, white noise, or air conditioning sounds.
2. **Simple Implementation**: The algorithm is relatively simple to implement using Fast Fourier Transform (FFT) and inverse FFT (IFFT) operations, making it computationally efficient and easy to understand.
3. **Real-time Application**: With proper optimization, this technique can be adapted for real-time speech enhancement applications, such as hands-free telephony, hearing aids, or voice assistants.

Disadvantages

1. **Over-Subtraction Impact**: If the over-subtraction factor α is too large, it can remove parts of the speech signal along with the noise, resulting in a loss of speech clarity and naturalness.
2. **Reduced Speech Quality in Low SNR**: In environments with very low signal-to-noise ratio (SNR), the method may struggle to distinguish between noise and speech, leading to excessive suppression of speech components.
3. **Musical Noise Artifacts**: Spectral subtraction can introduce an artificial “musical noise,” which consists of random tonal artifacts that occur when the noise estimation is not perfect.

Conclusion

In this project, a basic noise reduction algorithm was successfully implemented using spectral subtraction. The noisy speech signal was processed by dividing it into overlapping frames, applying a Hamming window, and performing short-time Fourier transforms (STFT). By estimating the noise in each frame and subtracting it from the magnitude spectrum, the algorithm significantly reduced background noise while preserving the important characteristics of the speech signal.

The project demonstrates the effectiveness of spectral subtraction for speech enhancement in noisy environments. However, further refinement could improve the quality of the enhanced speech, such as adjusting the noise estimation method or incorporating more advanced techniques like Wiener filtering or deep learning-based noise suppression.

Overall, this project provides a solid foundation for real-time speech enhancement applications, particularly in situations like telecommunication systems, hearing aids, and voice-activated devices where noise reduction is critical. With additional optimizations and experimentation, this approach can be adapted to a variety of noise environments to improve the intelligibility and quality of speech signals.

THANK YOU